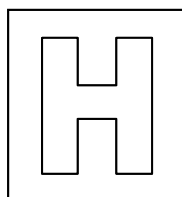


Candidate Name: _____

Class Adm No

--	--



2013 Preliminary Examination 2

Pre-University 3

H2 Biology

9648/03

20 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.
At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This question paper consists of 16 printed pages.

[Turn over

Answer **all** questions

1. The pink bollworm moth, *Pectinophora gossypiella*, is a pest of cotton crops. The size of its population can be reduced by releasing large numbers of sterile male moths into cotton fields. The sterile male moths mate with wild females from the cotton fields, but no offspring are produced.

Over a period of three years, 20 million genetically modified (GM) sterile male moths were released in the USA. Each insect contained a gene coding for a red fluorescent protein (DsRed) taken from a species of reef coral. The added DNA also included a promoter.

(a) Explain why, in gene technology:

- (i) genes for fluorescent proteins such as DsRed are now more commonly used as markers than are genes for antibiotic resistance

.....

 [2]

- (ii) a promoter needs to be included when transferring a gene from a coral into an insect.

.....

 [3]

- (b)** DsRed is visible at all stages of the life cycle of the moth, but the presence of the gene in a particular individual can be confirmed by genetic fingerprinting, using gel electrophoresis.

- (i)** Outline the principles of gel electrophoresis.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii)** Explain how the presence of the gene for DsRed in a moth can be confirmed once electrophoresis is complete.

.....

.....

.....

.....

..... [2]

- (c)** DsRed allows sterile male moths to be distinguished from wild moths when caught in an insect trap in a field of cotton plants.

Suggest why it is important to be sure whether a moth caught in such a trap is a released sterile male or a wild insect.

.....

.....

.....

..... [2]

- (d) The United States Department of Agriculture has ruled that the release of sterile males to control insect pest numbers is environmentally preferable to all other alternatives.

Suggest what information would be needed to determine whether the release of the sterile male moths, carrying the gene for DsRed, has a damaging effect on the environment.

.....

.....

.....

..... [2]

[Total: 15]

BLANK PAGE

2.

- (a) Plant responses to environmental changes are co-ordinated by plant growth substances (plant hormones).

Suggest why plants need to be able to respond to their environment.

.....

.....

..... [2]

- (b) The following investigation was carried out into the effects of plant growth substances on germination

- a large number of lettuce seeds was divided into eight equal batches
- each batch of seeds was placed on moist filter paper in a Petri dish and given a different treatment

The different treatments are shown in Table 2.1. Each tick represents one of the eight batches of seeds.

Table 2.1

	Treatment	concentration of gibberellin (mol dm^{-3})			
		0.00	0.05	0.50	5.00
A	Water	✓	✓	✓	✓
B	Absciscic acid	✓	✓	✓	✓

The batches of seeds were left to germinate at 25°C in identical conditions and the percentage germination was calculated

Fig. 2.1 shows the results of this investigation

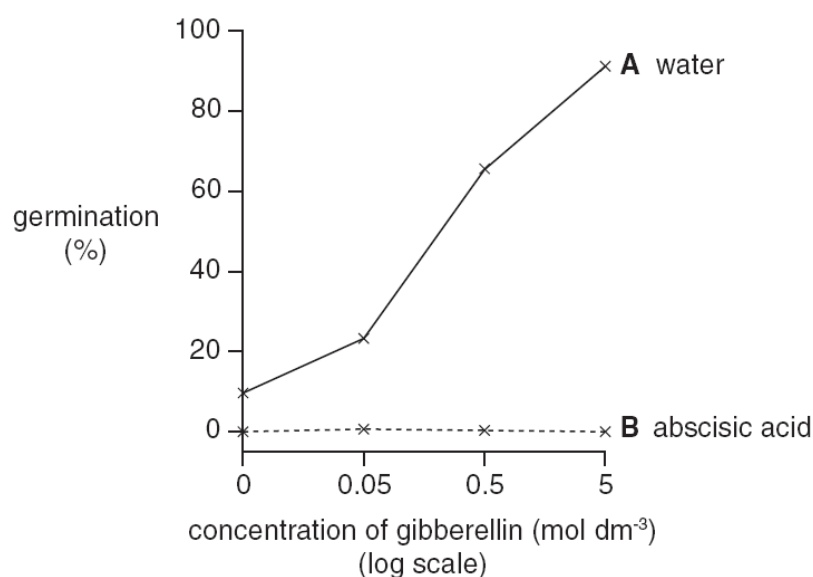


Fig. 2.1

- (i) Describe, with reference to Fig. 2.1, the effects of the plant growth substances on the germination of lettuce seeds.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii) Explain why all the lettuce seeds were kept at 25 °C.

.....

.....

.....

..... [2]

- (iii) State three variables, other than temperature, that needed to be controlled in the investigation.

1

2

3 [3]

- (c) Propose two commercial uses of plant growth substances.

.....

.....

.....

..... [2]

[Total: 13]

3. Genetic information in humans can be obtained by DNA profiling.

In DNA profiling, the polymerase chain reaction is used by a scientist to amplify a particular sequence of DNA.

(a) Briefly describe the steps of this technique.

.....

.....

.....

.....

.....

.....

.....

..... [4]

Scientists investigating the performance of athletes found that one gene contributing to the performance of sprinters is the *ACTN3* gene. There are two alleles of the gene, the 577R allele and the 577X allele. The 577X allele codes for a very short protein fragment in muscle fibres due to a stop codon mutation.

Table 3.1 summarises the athletic potential for the three possible genotypes for the *ACTN3* gene.

Table. 3.1

ACTN3 genotype	Athletic potential
577R / 577R	outstanding sprinter
577R / 577X	good sprinter or long-distance runner
577X / 577X	very good long-distance runner

A scientist tested sprinters to see if they possessed the 577R allele. Samples were obtained from athletes' muscle fibres. A standard containing proteins of the same lengths as the proteins coded for by both alleles 577X and 577R was used as a comparison. The standard and the samples were exposed to gel electrophoresis. In gel electrophoresis, protein molecules separate in the same way as DNA molecules.

The result for the standard is shown in Fig. 3.1.



Fig. 3.1

- (b)** On Fig. 3.1, draw the bands expected for an outstanding sprinter and for a good sprinter.

Explain why you have placed the bands in these positions.

.....

.....

.....

.....

.....

.....

.....

..... [4]

A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 3.2.

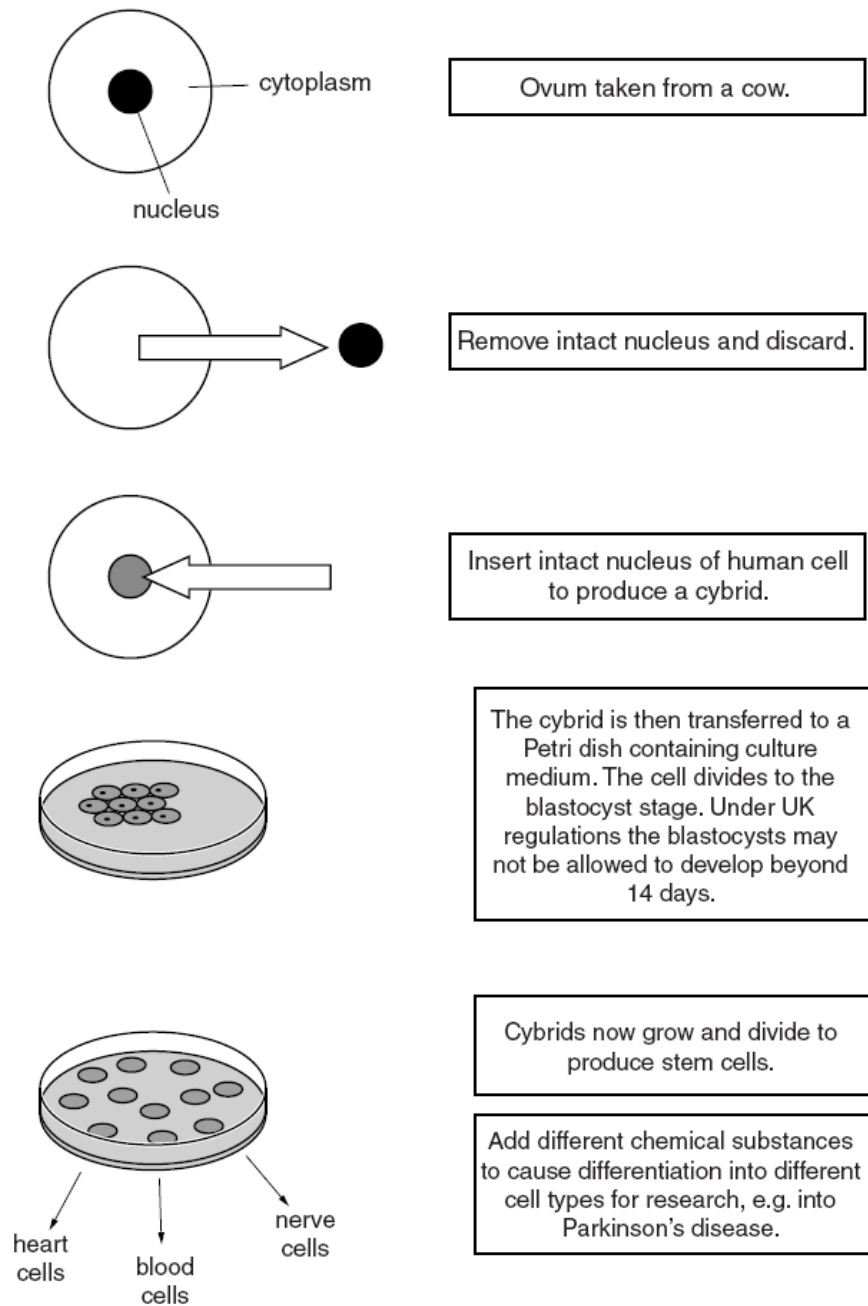


Fig. 3.2

- (c) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

.....

.....

.....

..... [1]

- (d) When the Human Fertilisation and Embryology Bill was considered by the UK Parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you agree or disagree that this is unethical and explain why you reached this decision.

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 12]

4. The enzyme urease is a catalyst of the hydrolysis of urea in solution, forming ammonia and carbon dioxide, for example in the breakdown of urea in soils by microorganisms.

You are required to plan an investigation to compare the activity of urease free in solution and urease immobilised in alginate beads.

As the reaction proceeds, the ammonia released dissolves, causing the pH to increase.

You are provided with the following equipment which you may use or not in your plan, as you wish. You may not use any additional equipment in your plan.

- an unlimited supply of calcium alginate beads all of uniform size prepared with a 50 g dm^{-3} urease solution (you may call this immobilised urease)
- an unlimited volume of 50 g dm^{-3} urease solution (you may call this free urease)
- an unlimited volume of 1.0 mol dm^{-3} urea solution
- an unlimited volume of distilled water
- beakers and flasks of different sizes
- stop watch or electronic timer
- broad and narrow range pH papers and liquids with appropriate colour charts, pH probes and meters
- colorimeter and tubes/cuvettes
- thermometer
- thermostatically-controlled water baths
- graduated pipettes and pipette fillers
- filter funnels
- syringes
- glass rods for stirring
- test-tubes and boiling tubes
- test-tube racks

Your plan should include a clear statement of the hypothesis or prediction and should identify key variables.

Your plan should be written in clear scientific language and should be illustrated with a diagram.

Your plan should give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible. A brief risk assessment should be included.

[Total: 12]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

This image shows a full page of white paper with horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

- (a)** Describe the genetic diseases SCID (severe combined immunodeficiency) and cystic fibrosis. [12]
- (b)** Discuss the factors which prevent gene therapy from becoming an effective treatment. [8]

[Total: 20]